Abstract

We study whether deep learning models can extract economically meaningful short-horizon signals from high-frequency cryptocurrency limit order book (LOB) data, and whether PCA compression and cross-asset features help. Using 1,030,551 one-second Bitcoin–Ethereum LOB snapshots from 7–19 April 2021, we build a single feature pipeline (raw or PCA-compressed LOB depth, cross-asset market state and six microstructure factors). On it we evaluate a Residual CNN, a bidirectional LSTM, a Transformer and a LightGBM baseline under one protocol: a purged train/validation/test split, trading rules chosen on validation only, and a non-overlapping backtest with a 1.5 bp fee.

All models show a weak but consistent directional signal (directional F1 0.44–0.47), although their accuracy is no better than always predicting the majority Flat class. Trading every prediction loses 39–49%, and confidence thresholds selected out of sample leave almost no trades. A hysteresis rule with separate entry and exit levels makes the signal profitable under the backtest's assumptions. Over six walk-forward test days, a seed-averaged CNN earns +3.1% to +10.0% per day against a buy-and-hold mean of −1.6%. However, this profit disappears when execution is delayed by one to three seconds. The signal is real but decays within about a second, so we find no tradable edge at one-second resolution after costs. Longer prediction horizons (30–300 s) do not help, and the deep architectures add little directional skill over the LightGBM baseline.

3 Research Questions

This project addresses the following questions:

RQ1. Can deep learning architectures (CNN, LSTM, Transformer) extract directional signals from one-second cryptocurrency LOB data at a 10-second horizon, and do they outperform a non-sequential gradient-boosting baseline?

RQ2. Does PCA-based compression of raw LOB features improve prediction and trading performance compared with using the raw order-book features directly?

RQ3. Do market state variables (rolling volatility, momentum, cross-asset ETH signals) and classic microstructure factors improve short-horizon prediction?

RQ4. Can the models generate returns after a 1.5 bp transaction cost when every trading parameter is chosen out of sample, and does any such return survive realistic execution delays?

RQ5. How do the prediction horizon (10–300 s) and the trading rule (argmax, confidence threshold, hysteresis) affect economic performance?

4 Data

4.1 Source and Coverage

We use the *High-Frequency Crypto Limit Order Book Data* dataset on Kaggle[1], which provides one-second LOB snapshots for BTC, ETH and ADA. We use the full BTC and ETH files. Timestamps are floored to the second (keeping the last snapshot within a second), and the two assets are inner-joined on that second. After dropping rows whose rolling features are not yet defined, 1,030,551 aligned snapshots remain, covering 7 April 2021 11:34 UTC to 19 April 2021 09:54 UTC (about 12 days).

[1] `https://www.kaggle.com/datasets/martinsn/high-frequency-crypto-limit-order-book-data`

Each BTC snapshot records 15 levels on each side: bid distances $\{d_i^b\}_{i=0}^{14}$, ask distances $\{d_i^a\}_{i=0}^{14}$, bid notionals $\{v_i^b\}_{i=0}^{14}$ and ask notionals $\{v_i^a\}_{i=0}^{14}$, i.e. 60 LOB features. Here *distance* is the relative offset of level i from the mid-price and *notional* is the dollar value resting at that level. From ETH we use the mid-price, spread and level-0 notionals.

4.2 Label Construction

For a horizon of h seconds the target is the forward mid-price return

$$r_t^{(h)} = \frac{p_{t+h} - p_t}{p_t}, \tag{1}$$

where p_{t+h} is looked up *by timestamp*, not by row offset. The feed has gaps, so "h rows later" is not always "h seconds later". Rows whose future timestamp is missing are never used as targets. Labels are three-class:

$$y_t = \begin{cases} \text{Up} & \text{if } r_t^{(h)} > +\delta_h, \\ \text{Down} & \text{if } r_t^{(h)} < -\delta_h, \\ \text{Flat} & \text{otherwise.} \end{cases} \tag{2}$$

The main horizon is $h = 10\,\text{s}$ with $\delta_{10} = 1.5 \times 10^{-4}$, which gives about 21% Up, 57% Flat and 21% Down. For the horizon study (Section 7.4) we also build labels for $h \in \{30, 60, 300\}\,\text{s}$ with $\delta_h = 3 \times 10^{-4}$, the round-trip cost, so that an Up label marks a move that is still profitable after paying to enter and exit. The share of Flat labels falls with the horizon (43.1%, 35.9%, 24.4% and 10.7% of the test set at 10, 30, 60 and 300 s).

4.3 Train–Validation–Test Split

We use a strictly chronological 70/15/15 split: training (7 April 11:34 to about 15 April 20:00), validation (to about 17 April 15:00) and test (to 19 April 09:54). The validation split is used for early stopping and for choosing every trading parameter. The test split is evaluated once.

Instead of a fixed embargo of 10 rows, we *purge* by time. A row is a valid target only if its whole label window $[t, t+h]$ lies inside its own split. No training label is then computed from validation or test prices, and no validation label from test prices, for any horizon h. Input windows may reach back across a split boundary. This is harmless, because all features are backward-looking.

To check that results do not depend on this single test window, Section 7.5 also uses *walk-forward* folds. The data are cut into 24-hour blocks from the first timestamp. Fold k trains on blocks $k, \ldots, k+4$, validates on block $k+5$ and tests on block $k+6$, giving six disjoint test days (13–18 April). Each fold has about 432k training, 86k validation and 86k test targets.

5 Methodology

5.1 Feature Engineering Pipeline

A single pipeline produces one feature table that all models read, so every model sees identical features, labels and splits.

5.1.1 Stage 1: Raw LOB Features

The 60 BTC LOB features (distances and notionals at 15 levels per side) form the base input and preserve the full depth profile of the book.

5.1.2 Stage 2: Market State Features

Fifteen features capture BTC and ETH short-term dynamics and their interaction:

- BTC: spread, level-0 order imbalance, rolling volatility and momentum of log returns (10 s and 60 s windows);
- ETH: the same six features from the ETH book;
- Cross-asset: BTC–ETH momentum difference (10 s, 60 s) and BTC/ETH volatility ratio (10 s).

5.1.3 Stage 3: Classic Microstructure Factors

Six factors from the microstructure literature yield 13 columns:

1. **Queue imbalance** [**?**]: exponentially weighted imbalance over five levels, $\mathrm{QI}_t = \sum_{i=0}^{4} w_i \frac{v_i^b - v_i^a}{v_i^b + v_i^a}$ with $w_i \propto e^{-0.5i}$, $\sum_i w_i = 1$.
2. **Order flow imbalance (OFI)** [**?**]: $\mathrm{OFI}_t = e_t^b - e_t^a$ with $e_t^b = v_{0,t}^b \mathbb{1}[P_t^b \geq P_{t-1}^b] - v_{0,t-1}^b \mathbb{1}[P_t^b \leq P_{t-1}^b]$ and e_t^a defined symmetrically, plus 10 s and 60 s running sums. Because the data store distances rather than prices, the best bid and ask prices are rebuilt as $P_t^b = p_t(1 + d_{0,t}^b)$ and $P_t^a = p_t(1 + d_{0,t}^a)$ and rounded to the \$0.01 tick before comparison. Comparing distances directly would misclassify every mid-price move as a queue change.
3. **Kyle's λ** [**?**]: rolling 60 s regression slope of the log return on OFI, $\widehat{\mathrm{Cov}}(r, \mathrm{OFI}) / \widehat{\mathrm{Var}}(\mathrm{OFI})$.
4. **Depth ratio** [**?**]: ask-to-bid notional ratio over the top 10 levels, the corresponding imbalance, and a 10 s rolling mean of the ratio.
5. **Tick sign** [**?**]: sign of the most recent non-zero mid-price return, forward-filled, plus 10 s and 60 s running sums.
6. **Amihud illiquidity** [**?**]: $|r_t|$ divided by the top-5-level book notional (book depth stands in for traded volume), plus its 60 s rolling mean.

5.1.4 Stage 4: Normalization

Every feature is standardised with a backward-looking rolling z-score over the last 3,600 seconds (minimum 60 observations),

$$\tilde{x}_{t,j} = \mathrm{clip}\left(\frac{x_{t,j} - \hat{\mu}_{t,j}}{\hat{\sigma}_{t,j}},\ -10,\ 10 \right), \tag{3}$$

where $\hat{\mu}_{t,j}$ and $\hat{\sigma}_{t,j}$ use observations $t - 3599, \ldots, t$ only, so no future information enters. Two details matter. First, no additive constant is used in the denominator, and windows with $\hat{\sigma}_{t,j} = 0$ map to 0. Amihud and Kyle's λ are $\mathcal{O}(10^{-10})$, and an $\varepsilon = 10^{-8}$ (as in an earlier version) squashes them to zero. Second, values are clipped at ± 10 because near-constant columns, such as a spread pinned at one tick, otherwise produce extreme values whenever they move.

5.1.5 Stage 5: PCA Compression

A `StandardScaler` and PCA are fitted on the 60 normalized LOB features of the *training rows only* and applied to all rows. About 30 components are needed for 90% of the variance. We keep the components with eigenvalue above one (Kaiser criterion), capped at 10, which retain about 61% of the variance. In the walk-forward evaluation, the scaler and PCA are refitted on each fold's training days.

5.1.6 Feature Sets

Three feature sets are used throughout, with the same definition for every model (Table 1).

Table 1: Feature sets. All are columns of one feature table with 1,030,551 rows.

Name	Short name	Feature groups	Features
Full	`full`	PCA(10) + Market(15) + Classic(13)	38
No PCA	`no_pca`	Raw LOB(60) + Market(15) + Classic(13)	88
No Market	`no_market`	PCA(10) + Classic(13)	23

5.2 Model Architectures

[architecture bullet lists unchanged]

All models output a probability distribution $\hat{\mathbf{p}}_t = (\hat{p}_\downarrow, \hat{p}_0, \hat{p}_\uparrow)$ and are trained with class-weighted cross-entropy,

$$\mathcal{L} = -\sum_c w_c \mathbb{1}[y = c] \log \hat{p}_c, \qquad w_c = \frac{N}{3N_c}, \tag{4}$$

where N_c is the number of training targets in class c. All neural networks use AdamW, gradient clipping at norm 1.0, shuffled mini-batches, and *early stopping on validation loss*. The weights of the best validation epoch are restored.

5.2.1 Residual CNN

Key hyperparameters: 64 filters, $T = 10$, dropout 0.3, learning rate 10^{-4}, weight decay 10^{-2}, cosine annealing over at most 50 epochs, patience 10, batch size 256. Total parameters: 118,851 for $F = 38$.

5.2.2 Bidirectional LSTM

Key hyperparameters: $T = 10$, hidden size 32 per direction, 2 layers, dropout 0.4, learning rate 10^{-4}, weight decay 2×10^{-3}, cosine annealing over at most 40 epochs, patience 6, batch size 256.

5.2.3 Transformer

Key hyperparameters: $T = 30$, $d_{\text{model}} = 64$, 4 heads, 3 pre-norm layers, feed-forward width 128, dropout 0.1, OneCycleLR with maximum learning rate 10^{-3} (10% warm-up, cosine decay), weight decay 10^{-4}, at most 30 epochs, patience 6, batch size 512. Total parameters: 105,219 for $F = 38$.

5.2.4 LightGBM Baseline

To measure what the sequence architectures add, we train a gradient-boosted tree classifier (LightGBM) on the features of the *current second only*, with no look-back window. It uses up to 2,000 trees, learning rate 0.05, 31 leaves, a minimum of 200 samples per leaf, 80% row and column subsampling, balanced class weights, and early stopping on validation multi-class log-loss (50 rounds).

5.3 Backtesting Framework

5.3.1 Positions, Costs and Returns

A trading rule maps each prediction to a position $s_t \in \{-1, 0, +1\}$ (short, flat, long), using the full account value without leverage. The position is held for exactly h seconds and the net return of the step is

$$r_t^{\text{net}} = s_t \, r_t^{(h)} - c \, |s_t - s_{t-1}|, \tag{5}$$

with $c = 1.5\,\text{bp}$ per unit of position change. A reversal from -1 to $+1$ therefore pays $2c$, and the first entry is also charged. Wealth compounds as $W_T = \prod_t (1 + r_t^{\text{net}})$. Execution is at the mid-price of the snapshot from which the features were computed. Section 7.6 tests this assumption by delaying execution.

5.3.2 Non-Overlapping Steps

Predictions are made every second, but each label covers h seconds. Using every prediction would count each price move up to h times. We therefore keep a subset of target rows such that each is at least h seconds after the previous one. The positions are taken at these rows and earn non-overlapping returns. For $h = 10$ the test split has 15,457 such steps. Sharpe ratios are annualised with $\sqrt{365 \times 24 \times 3600/h}$ (≈ 1776 at $h = 10\,\text{s}$). These values are only comparable across models in this study, not with daily-frequency strategies.

5.3.3 Trading Rules

Let $\hat{p}_\downarrow$ and $\hat{p}_\uparrow$ be the predicted probabilities of Down and Up.

- **Argmax**: $s_t = \arg\max_c \hat{p}_c - 1$, i.e. trade every prediction.
- **Threshold**: $s_t = +1$ if $\hat{p}_\uparrow > \theta$, $s_t = -1$ if $\hat{p}_\downarrow > \theta$, otherwise 0.
- **Hysteresis**: open a long (short) position when $\hat{p}_\uparrow$ ($\hat{p}_\downarrow$) exceeds an *enter* level θ_{in} and is the larger of the two directional probabilities. Keep the position while its probability stays at or above an *exit* level $\theta_{\text{out}} < \theta_{\text{in}}$, and reverse directly when the opposite probability exceeds θ_{in}. The lower exit level avoids paying to close and reopen a position on every small dip in confidence.

5.3.4 Out-of-Sample Parameter Selection

θ is swept over 23 values in $[0.35, 0.90]$, and $(\theta_{\text{in}}, \theta_{\text{out}})$ over $\theta_{\text{in}} \in \{0.40, 0.45, \ldots, 0.90\}$ and $\theta_{\text{out}} \in \{0, 0.05, \ldots, \theta_{\text{in}} - 0.05\}$. Both sweeps are run on the *validation* split, and the configuration with the highest validation Sharpe among those with at least 20 position changes is applied once to the test split. An earlier version of this study selected θ on the test split, which inflates test performance.

5.3.5 Performance Metrics

Statistical: accuracy, per-class precision, recall and F1, and directional F1,

$$\mathrm{DirF1} = \tfrac{1}{2}\left(\mathrm{F1}_{\mathrm{Down}} + \mathrm{F1}_{\mathrm{Up}}\right), \tag{6}$$

which ignores the dominant Flat class. Accuracy is reported next to the accuracy of always predicting Flat. *Economic:* cumulative net return, annualised Sharpe ratio, maximum drawdown, win rate on active steps, and turnover $\sum_t |s_t - s_{t-1}|$. *Robustness:* the test return over neighbouring threshold values, the split of P&L into long and short positions and by calendar day, results across walk-forward folds and seeds, and the test return when execution is delayed by $d \in \{1, 2, 3, 5\}$ seconds.

6 Experimental Setup

Neural networks are trained on Google Colab with an NVIDIA Tesla T4 GPU, and LightGBM on the Colab CPU. All code (feature pipeline, models, backtest and the scripts that produce every table) is shared across models, so differences in results come from the models, not from preprocessing or evaluation.

Experiments on the main split use seed 42. They cover all four models and three feature sets at $h = 10$ s, and the Full feature set at $h \in \{30, 60, 300\}$ s (LightGBM: all feature sets). The walk-forward evaluation trains the CNN with seeds 42, 43 and 44 plus a seed-averaged ensemble, and LightGBM with seed 42, on six folds at $h = 10$ s. Development was first done on the first ≈1.2 days of data (100,000 rows) and then scaled to the full dataset. That development subset lies entirely inside the main training split.

Table 2: Hyperparameters by model. All neural networks use AdamW, gradient clipping at 1.0, shuffled batches, class-weighted cross-entropy and early stopping on validation loss.

	CNN	BiLSTM	Transformer	LightGBM
Input window T	10	10	30	1
Width	64 / 128 filters	32×2	$d_{\mathrm{model}} = 64$	31 leaves
Depth	2 residual blocks	2 layers	3 layers	$\leq$ 2,000 trees
Dropout	0.3	0.4	0.1	–
Learning rate	10^{-4}	10^{-4}	10^{-3} (max)	0.05
Schedule	cosine	cosine	OneCycle	–
Weight decay	10^{-2}	2×10^{-3}	10^{-4}	$\lambda_2 = 1$
Batch size	256	256	512	–
Max epochs	50	40	30	–
Early stopping	val loss, 10	val loss, 6	val loss, 6	val log-loss, 50
Parameters ($F = 38$)	118,851	45,827	105,219	–

7 Results and Discussion

7.1 Evaluation Protocol

All results use the shared pipeline of Section 5.1, the purged train/validation/test split of Section 4.3 (test: about 17 April 15:00 to 19 April 09:54 UTC, 154,544 targets at $h = 10$ s), and the backtest

of Section 5.3. Early stopping and every trading parameter (the threshold θ and the hysteresis levels) are chosen on the validation split, and the test split is evaluated once. Execution is at the mid-price of the snapshot the features come from; Section 7.6 relaxes this assumption.

7.2 Classification Performance

Table 3 reports test classification metrics at the 10-second horizon. All four models reach 41–44% accuracy, which is *no better than always predicting Flat* (43.1% of test labels). Accuracy is therefore not informative here. Directional F1, the macro F1 over the Down and Up classes, does show a weak but consistent directional signal of 0.44–0.47. The Transformer has the highest directional F1 (0.466), but the gap to the other models is small: LightGBM, which sees only the current second, reaches 0.465. The sequence architectures add little classification skill beyond the hand-crafted features.

Table 3: Test classification metrics, 10-second horizon, main split (154,544 test targets). DirF1 = macro F1 over Down and Up. Always predicting Flat gives an accuracy of 0.431.

	Full (38F)		No PCA (88F)		No Market (23F)	
Model	Acc	DirF1	Acc	DirF1	Acc	DirF1
Residual CNN	0.445	0.454	0.444	0.448	0.443	0.437
BiLSTM	0.434	0.447	0.434	0.449	0.436	0.445
Transformer	0.437	**0.466**	0.434	0.464	0.436	0.456
LightGBM baseline	0.409	0.465	0.408	0.461	0.413	0.444

Across feature sets, removing the market-state block (No Market) lowers directional F1 for every model (by 0.002–0.020), while replacing PCA factors with the raw 60 LOB columns (No PCA) changes it by less than 0.007. Cross-asset features help slightly; PCA compression neither helps nor hurts.

7.3 Trading Performance on the Main Split

Table 4 reports the test backtest at the 10-second horizon. Buy-and-hold loses 6.5% over the test window, which contains the 18 April sell-off (maximum drawdown −14.8%).

Argmax. Trading every prediction loses 39–49% for every model and feature set. Even with a non-trivial directional F1, the expected move per 10-second step is far smaller than the 1.5 bp paid on each position change.

Threshold. A confidence threshold chosen on validation ($\theta = 0.55$–0.85) almost stops the models from trading: 20–280 position changes over 1.8 days, with test returns between −0.7% and +0.7%. With so few trades these results are not distinguishable from zero. In particular, the earlier draft's CNN result of +2.7% (Sharpe 31) does not survive once θ is selected out of sample.

Hysteresis. Separating the entry and exit levels changes the picture. The model now stays in a position through small dips in confidence instead of paying to close and reopen it. Five of the six hysteresis runs turn positive. The CNN earns +26.8% (Sharpe 62, maximum drawdown −6.4%). On this split, the result is not a single lucky configuration. Both long positions (+15.1%)

Table 4: Test returns at the 10-second horizon (15,457 non-overlapping steps, 1.5 bp fee). θ and the hysteresis levels are chosen on validation. Turnover = total $\sum_t |q_t - q_{t-1}|$. Buy-and-hold: −6.52%.

		Argmax	Threshold		Hysteresis	
Model	Features	Return	Return	Turnover	Return	Turnover
Residual CNN	Full	−40.1%	+0.72%	28	**+26.8%**	2,249
Residual CNN	No PCA	−40.3%	−0.35%	42	–	–
Residual CNN	No Market	−42.2%	−0.71%	62	–	–
BiLSTM	Full	−48.6%	−0.16%	96	+4.5%	428
BiLSTM	No PCA	−46.9%	+0.27%	22	–	–
BiLSTM	No Market	−46.5%	−0.12%	54	–	–
Transformer	Full	−42.8%	+0.32%	152	+3.5%	284
Transformer	No PCA	−39.7%	−0.06%	24	–	–
Transformer	No Market	−42.5%	+0.05%	30	–	–
LightGBM	Full	−39.3%	+0.08%	20	+6.3%	974
LightGBM	No PCA	−42.6%	−0.21%	280	+17.1%	1,922
LightGBM	No Market	−42.5%	−0.34%	114	−0.9%	1,039

and short positions (+22.9%) contribute, before −14.1% of fees. Each of the three test days is positive (+3.6%, +16.9%, +3.5%). Every neighbouring (enter, exit) pair within ±0.05 also stays positive (+14.1% to +49.0%). The LSTM (+3.0% to +14.7% over the same neighbourhood) and the Transformer (+0.0% to +11.9%) are positive on every test day as well, with far less trading.

7.4 Effect of the Prediction Horizon

A natural way to beat the fee is to predict further ahead, since price moves grow with the horizon while the fee does not. We rebuilt the labels for $h \in \{30, 60, 300\}$ seconds, with the Up/Down threshold raised to the round-trip cost of 3 bp, and retrained every model on the Full feature set (Table 5).

Table 5: Directional F1 and hysteresis test return by prediction horizon (Full feature set, main split). "Flat" = share of Flat labels in the test set, which equals the accuracy of always predicting Flat. Buy-and-hold ≈ −6.6% at every horizon.

		CNN		BiLSTM		Transformer		LightGBM	
h	Flat	DirF1	Hyst.	DirF1	Hyst.	DirF1	Hyst.	DirF1	Hyst.
10 s	0.431	0.454	+26.8%	0.447	+4.5%	0.466	+3.5%	0.465	+6.3%
30 s	0.359	0.413	+0.1%	0.426	+0.7%	0.437	+1.0%	0.446	+13.3%
60 s	0.244	0.436	+8.3%	0.445	+4.0%	0.416	−0.3%	0.466	+8.6%
300 s	0.107	0.420	−4.4%	0.447	−3.9%	0.372	−12.3%	0.476	−2.4%

Longer horizons do not help. Accuracy rises above the Flat baseline only because Flat labels become rare, and directional F1 stays at 0.37–0.48 at every horizon. The apparent gains at 30 and 60 seconds come mostly from short positions during the 18 April sell-off, and they are fragile. For LightGBM at 30 seconds, for example, the test return over the neighbouring (enter, exit) pairs ranges from −12.1% to +14.9%. At 300 seconds every hysteresis run loses money. LightGBM

matches or exceeds the deep models' directional F1 at every horizon.

7.5 Walk-Forward Evaluation

The main split has a single 1.8-day test window. Many configurations were compared on it, so its best result is optimistically selected. To test whether the 10-second hysteresis result generalises, we re-ran the CNN and LightGBM on rolling folds. Each fold trains on 5 days, validates on the next day and tests on the day after, and the window then moves forward one day, giving six disjoint test days from 13 to 18 April. Within each fold, the PCA basis is refitted on that fold's training days, the hysteresis levels are chosen on its validation day, and the CNN is trained with three seeds plus a seed-averaged ensemble. The first four test days were not used to choose this configuration, so they are genuinely out of sample.

Table 6: Walk-forward test-day returns (10-second horizon, hysteresis, 1.5 bp fee). Each column is one fold; a "day" starts at 11:34 UTC. CNN single seeds: mean of the three seeds on each day (individual runs range from +1.0% to +35.3%).

Test day	04-13	04-14	04-15	04-16	04-17	04-18	Mean
CNN, 3-seed ensemble	+4.2%	+3.1%	+9.8%	+9.0%	+6.2%	+10.0%	+7.0%
CNN, single-seed mean	+2.9%	+2.7%	+3.6%	+6.7%	+20.6%	+9.8%	+7.7%
LightGBM, No PCA	+3.2%	+8.9%	+7.9%	+5.9%	−0.8%	+11.3%	+6.1%
LightGBM, Full	−2.2%	+1.0%	+12.4%	+9.3%	−2.7%	+10.8%	+4.8%
Buy-and-hold	+2.3%	−3.0%	−3.0%	+0.7%	−10.9%	+4.5%	−1.6%

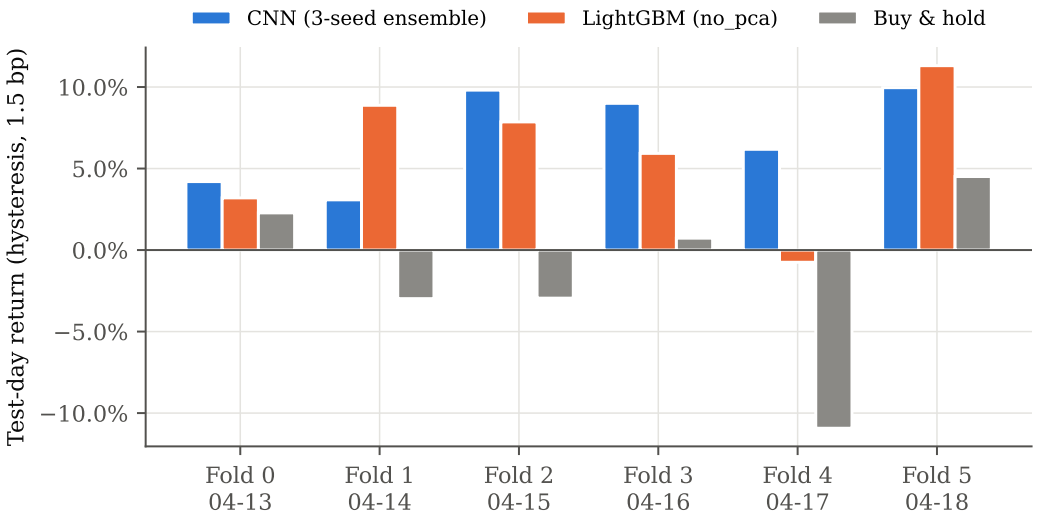


Figure 1: Walk-forward test-day returns of the hysteresis strategy against buy-and-hold.

Table 6 and Figure 1 show that the result is remarkably stable *under the backtest's assumptions.* All 18 single-seed CNN runs and all six ensemble days are positive, including the four test days before the sell-off, which were not used to choose the configuration. Compounded over the six days, the CNN ensemble returns +50.1% against −9.7% for buy-and-hold. Seed averaging lowers the variance: single seeds range from +1.0% to +35.3%, the ensemble from +3.1% to +10.0%.

LightGBM is profitable on 5 of 6 days with No PCA features and on 4 of 6 with Full features. Directional F1 stays at 0.41–0.47 in every fold, the same level as on the main split.

7.6 Reality Checks: Execution Delay and Data Alignment

Execution delay. The backtest assumes we trade at the mid-price of the very snapshot from which the features were computed. To test this, we keep every trained model and every validation-chosen threshold fixed and decide the position for the step starting at second t from the prediction made at second $t - d$. This uses only the saved predictions, so no retraining is involved. Figure 2 and Table 7 show the result.

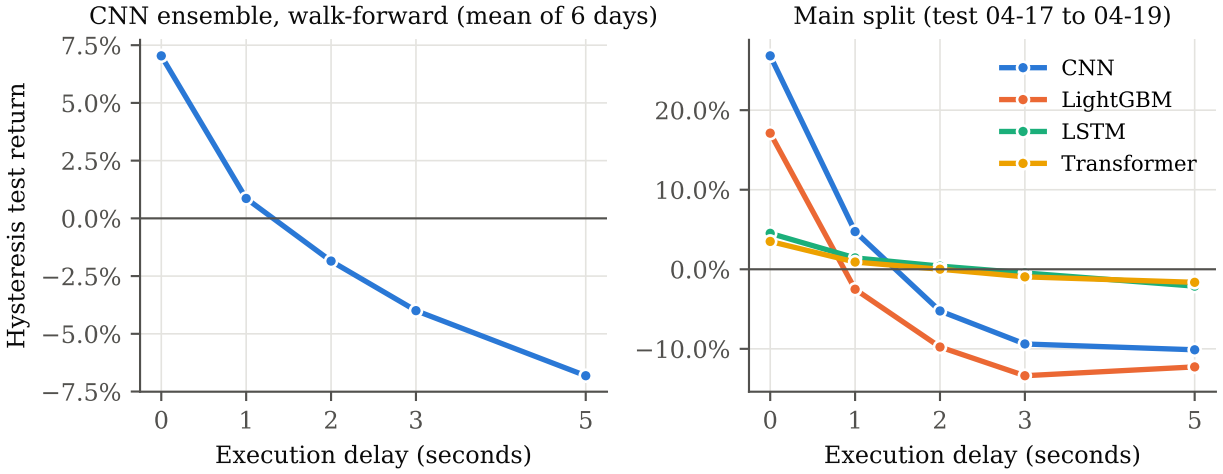


Figure 2: Hysteresis test return as a function of execution delay. Left: CNN ensemble, mean over the six walk-forward test days. Right: all models on the main split.

Table 7: Hysteresis test return with execution delayed by d seconds (10-second horizon, same models and thresholds). Walk-forward rows: mean over six test days (share of positive days in brackets).

	$d = 0$	$d = 1\,\text{s}$	$d = 2\,\text{s}$	$d = 3\,\text{s}$	$d = 5\,\text{s}$
Walk-forward (mean per test day)					
CNN ensemble	+7.0% (6/6)	+0.9% (4/6)	−1.9% (3/6)	−4.0% (2/6)	−6.8% (0/6)
CNN single seeds	+7.7%	+2.3%	−1.0%	−2.9%	−4.9%
LightGBM (both sets)	+4.9%	−3.6%	−4.2%	−6.3%	−9.0%
Main split (test 04-17 to 04-19)					
CNN	+26.8%	+4.7%	−5.2%	−9.4%	−10.1%
BiLSTM	+4.5%	+1.4%	+0.4%	−0.5%	−2.1%
Transformer	+3.5%	+0.9%	+0.0%	−1.0%	−1.6%
LightGBM, No PCA	+17.1%	−2.5%	−9.8%	−13.4%	−12.3%

For every model, most of the profit disappears with a one-second delay, and it turns negative at two to three seconds. The models do anticipate the next move of the mid-price, but that move happens within roughly one second of the snapshot. The LSTM and Transformer lose less when delayed only because their validation-chosen entry levels are higher, so they trade less. Their profit

decays in the same way.

BTC–ETH timestamp alignment. BTC and ETH snapshots are joined on the floored second. Within a matched second (first 300,000 snapshots), the ETH snapshot is recorded *later* than the BTC snapshot 24.5% of the time, by up to 0.74 s (19.7% of seconds are more than 0.1 s later). In those seconds, the ETH-based market features contain information from slightly after the BTC mid-price they are compared with, which is a small look-ahead in every feature set that includes market features. A delay of one second or more removes this leak as well, which is a further reason to treat the delayed numbers as the realistic ones.

7.7 Discussion

The results support three conclusions.

1. **The models learn genuine but very short-lived order-book information.** Directional F1 of 0.41–0.47 is stable across architectures, feature sets, seeds and six walk-forward folds. Under instantaneous execution, a hysteresis rule turns this signal into profits that are consistent across days and robust to the choice of thresholds.

2. **The signal is not tradable at 1-second resolution after a 1.5 bp fee.** Its value is concentrated in the first second after each snapshot. With any realistic latency, no model, feature set, horizon or trading rule in this study has a positive expected return. Capturing it would require sub-second execution at the mid-price, with no queueing or adverse selection, which this data cannot evaluate.

3. **Architecture matters less than evaluation design.** The three deep models and a LightGBM baseline that sees only the current second reach similar directional F1. The choices with the largest effect on reported returns are methodological: selecting thresholds out of sample (which removes the earlier +2.7% CNN result), separating entry from exit, and modelling execution delay (which removes all remaining profit).

Limitations. The data covers only 12 days in April 2021, and all test days fall in one week that contains a single large sell-off. The backtest fills at the mid-price with a flat fee, ignores market impact and borrowing costs for short positions, and uses full-capital positions. The 1.5 bp fee is lower than retail taker fees at the time. Only the CNN and LightGBM were evaluated walk-forward; the LSTM and Transformer were checked for execution delay on the main split only.

8 Conclusion

We built a single, leakage-controlled pipeline for one-second BTC limit order book prediction and evaluated three deep architectures and a gradient-boosting baseline under the same out-of-sample protocol. Our answers to the research questions are as follows.

1. **Directional signal (RQ1).** All models learn a weak but consistent directional signal: directional F1 is 0.44–0.47 on the main split and 0.41–0.47 across six walk-forward folds. Accuracy (41–44%) is no better than always predicting Flat. The Transformer has the highest directional F1 (0.466). However, a LightGBM model that sees only the current second matches it (0.465), so the sequence architectures add little beyond the hand-crafted features.

2. **PCA and market features (RQ2, RQ3).** Replacing the 60 raw LOB columns by 10 principal components changes directional F1 by less than 0.007. Removing the market-state block lowers it for every model (by up to 0.020). Cross-asset information helps slightly, while PCA compression neither helps nor hurts.

3. **Profitability after costs (RQ4).** Trading every prediction loses 39–49% at 1.5 bp. Confidence thresholds chosen on validation leave almost no trades. In particular, the +2.7% CNN result of our earlier draft does not survive out-of-sample threshold selection. A hysteresis rule makes the signal profitable under instantaneous execution. The CNN earns +26.8% on the main test window, and a 3-seed CNN ensemble is positive on all six walk-forward test days (mean +7.0% per day against −1.6% for buy-and-hold). This profit, however, is concentrated in the first second after each snapshot. With a one-second execution delay the ensemble's mean daily return falls to +0.9%, and by three seconds every model's return is negative. We also found that ETH snapshots are up to 0.74 s later than the BTC snapshots they are joined with in 24.5% of seconds, a small look-ahead that any realistic delay removes.

4. **Horizon and trading rule (RQ5).** Longer horizons (30, 60, 300 s) do not improve directional skill. Their occasional profits come from the 18 April sell-off and are fragile to the choice of thresholds. The trading rule matters far more than the architecture: separating entry from exit turned five of the six 10 s runs from losses of 39–49% into gains under the backtest's assumptions.

Overall, the LOB contains genuine predictive information about the next mid-price move, but at one-second resolution that information decays faster than it can be traded after a 1.5 bp fee. The most consequential choices in this study were methodological: out-of-sample selection of trading parameters, walk-forward testing, and modelling execution delay.

Limitations and future work. The data covers 12 days in a single month, and all test days fall in one week that contains one large sell-off. The backtest fills at the mid-price with a flat fee and full-capital positions, ignoring queue position, adverse selection, market impact and borrowing costs for short positions. Only the CNN and LightGBM were evaluated walk-forward. Natural next steps are (i) sub-second or event-time data, to measure whether the decaying signal can be captured with realistic latency; (ii) explicit maker/taker execution models with queue position; (iii) using traded volume (the `buys`/`sells` fields) for order-flow and Amihud factors; and (iv) longer samples covering several market regimes.